

4. Analyzing ecological footprints reveals that
 - (A) Earth's carrying capacity would increase if per capita meat consumption increased.
 - (B) current demand by industrialized countries for resources is much smaller than the ecological footprint of those countries.
 - (C) it is not possible for technological improvements to increase Earth's carrying capacity for humans.
 - (D) the ecological footprint of the United States is large because per capita resource use is high.
5. Based on current growth rates, Earth's human population in 2019 will be closest to
 - (A) 2.5 million.
 - (C) 7.8 billion.
 - (B) 4.5 billion.
 - (D) 10.5 billion.

Levels 3-4: Applying/Analyzing

6. The observation that members of a population are uniformly distributed suggests that
 - (A) resources are distributed unevenly.
 - (B) the members of the population are competing for access to a resource.
 - (C) the members of the population are neither attracted to nor repelled by one another.
 - (D) the density of the population is low.
7. According to the logistic growth equation
$$\frac{dN}{dt} = rN \frac{(K - N)}{K}$$
 - (A) the number of individuals added per unit time is greatest when N is close to zero.
 - (B) the per capita population growth rate increases as N approaches K .
 - (C) population growth is zero when N equals K .
 - (D) the population grows exponentially when K is small.
8. During exponential growth, a population always
 - (A) has a constant per capita population growth rate.
 - (B) quickly reaches its carrying capacity.
 - (C) cycles through time.
 - (D) loses some individuals to emigration.
9. Which of the following statements about human populations in industrialized countries is correct?
 - (A) Birth rates and death rates are high.
 - (B) Average family size is relatively large.
 - (C) The population has undergone the demographic transition.
 - (D) The survivorship curve is Type II.

Levels 5-6: Evaluating/Creating

10. **INTERPRET THE DATA** To estimate which age cohort in a population of females produces the most female offspring, you need information about the number of offspring produced per capita within that cohort and the number of individuals alive in the cohort. Make this estimate for Belding's ground squirrels by multiplying the number of females alive at the start of the year (column 2 in Table 53.1) by the average number of female offspring produced per female (column 5 in Table 53.1). Draw a bar graph with female age in years on the x -axis (0–1, 1–2, and so on) and total number of female offspring produced for each age cohort on the y -axis. Which cohort of female Belding's ground squirrels produces the most female young?

11. **EVOLUTION CONNECTION** Contrast the selective pressures operating in high-density populations (those near the carrying capacity, K) versus low-density populations.
12. **SCIENTIFIC INQUIRY** You are testing the hypothesis that increased population density of a particular plant species increases the rate at which a pathogenic fungus infects the plant. Because the fungus causes visible scars on the leaves, you can easily determine whether a plant is infected. Design an experiment to test your hypothesis. Describe your experimental and control groups, how you would collect data, and what results you would see if your hypothesis is correct.
13. **SCIENCE, TECHNOLOGY, AND SOCIETY** Some people regard the rapid population growth of less industrialized countries as our most serious environmental problem. Others think that the population growth in industrialized countries, though smaller, is actually a greater environmental threat. What problems result from population growth in (a) less industrialized countries and (b) industrialized nations? Which do you think is a greater threat, and why?
14. **WRITE ABOUT A THEME: INTERACTIONS** In a short essay (100–150 words), identify the factor or factors in Figure 53.17 that you think may ultimately be most important for density-dependent population regulation in humans, and explain your reasoning.
15. **SYNTHESIZE YOUR KNOWLEDGE**



Locusts (grasshoppers in the family Acrididae) undergo cyclic population outbreaks, leading to massive swarms such as this one in the Canary Islands, off the west coast of Africa. Of the mechanisms of density-dependent regulation shown in Figure 53.17, choose the two that you think most apply to locust swarms, and explain why.

For selected answers, see Appendix A.